

6 Meiosis results in genetic variation in gametes.

(a) (i) Explain how meiosis causes this genetic variation.

(3)

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(ii) Give **two** differences in the arrangement of the DNA in a cell at the beginning of interphase and at the end of prophase I.

(2)

1

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2

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(b) The DNA content of two cells was compared.

Cell A was undergoing mitosis. The mass of DNA in the cell in prophase was 12 pg.

Cell B was undergoing meiosis. The mass of DNA in the cell in prophase I was 12 pg.

What would be the mass of DNA in each daughter cell produced at the end of mitosis and at the end of meiosis?

(1)



A



B



C



D

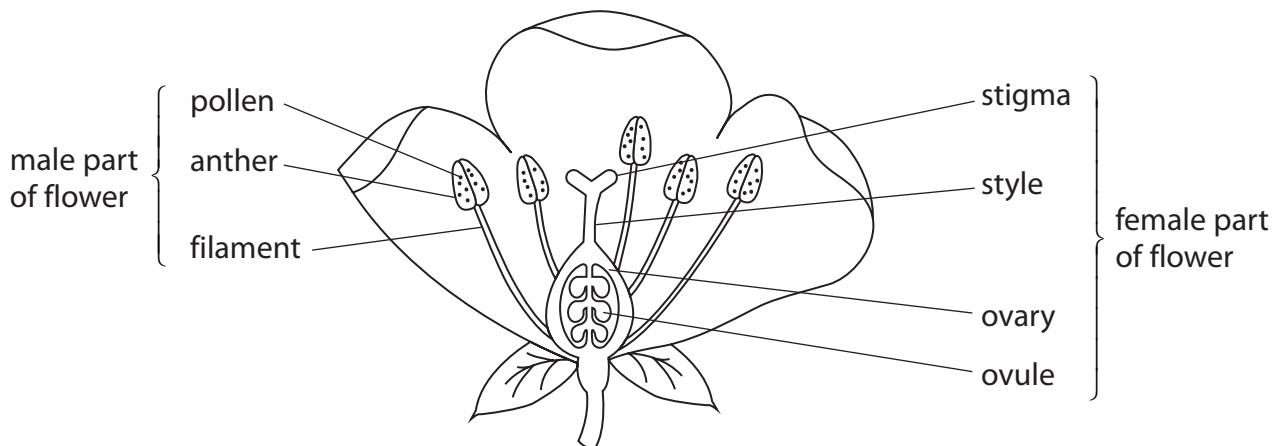
	Mass of DNA in a daughter cell produced from cell A at the end of mitosis / pg	Mass of DNA in a daughter cell produced from cell B at the end of meiosis / pg
	3	6
	6	3
	6	12
	12	6

*(c) Plants can produce seeds from either cross-fertilisation or self-fertilisation.

Self-fertilisation occurs when male nuclei from a flower fertilises an egg cell in a flower on the same plant.

Cross-fertilisation occurs when male nuclei from a flower on one plant fertilises an egg cell in a flower on a different plant of the same species.

The image shows the female and male parts of a flower.



Some plants have developed strategies to prevent self-fertilisation.

- Date palm plants have only male flowers or only female flowers.
- Some orchids release chemicals that prevent the germination of pollen produced by flowers of the same plant.

Some plants have developed strategies to produce seeds if cross-fertilisation has not occurred.

- The fireweed plant species from North America will self-fertilise if cross-fertilisation has not occurred.
- Unfertilised egg cells of a dandelion plant will develop into a seed.

Discuss the advantages and disadvantages of the strategies shown by these plants.

(6)

(Total for Question 6 = 12 marks)

